

A Framework for Emergent Time from Horn-Filling Dynamics

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Abstract

We propose a framework in which spacetime is a growing simplicial set and time is the process of filling its inner horns. Two local, non-deleting moves drive the growth of spacetime: a "sprout" attaches a fresh directed edge ending at a new vertex, and a "fill" completes an unfilled inner horn with a composite edge and its witnessing cell. Each move is a weak homotopy equivalence of the underlying space, so the homotopy type is conserved at every step. Since the geometric realization of the simplicial set forgets edge orientation, this conserved type cannot see any arrow of time; all physical content lives in the directed structure that realization forgets. The completed limit of the filling process is a quasicategory, in which every composition question has an answer while edges remain directed. Each move is an event that consumes cells created by earlier events, the dependency of moves forms a partial order, and a reference frame is a choice of linear extension of this order. The entropy of an event is the logarithm of the number of linear extensions of its causal past. This quantity measures how much of the past could have happened in a different order, and grows monotonically, which we prove; a toy implementation provides first numerical support. The arrow of time appears as a broken symmetry. The universe carries an arrow exactly when it is not isomorphic to its own edge-reversal, and pairing the universe with its mirror image restores the symmetry globally, in the spirit of CPT-symmetric cosmologies. We formulate a maximum-entropy inference problem whose solution, conjecturally of Gibbs form, would determine the rule from locality and a fixed update budget alone. We close with the open problems, foremost the recovery of the Lorentz group and of an area law for horizon entropy.

1 Introduction

Discrete approaches to quantum gravity, from Regge calculus [13] to causal dynamical triangulations [3], replace smooth spacetime with combinatorial data: simplices, gluings, and the lengths assigned to them. In Regge calculus the combinatorial scaffold is fixed with only edge lengths allowed to vary. The vertices, simplices, and gluings of the triangulation never change, so no new combinatorial structure is ever introduced. That is enough for classical gravity, where one solves the field equations on a triangulation that is given in advance. The discomfort begins one step later, when Feynman's path integral is applied to the evolution of the universe and the sum runs over all histories. Summing over Regge geometries means integrating over edge lengths, a continuous space in which many assignments describe the same geometry and no canonical measure presents itself. On the other hand, dynamical triangulations [8, 1, 2] escape this by inverting Regge's choice. The lengths are frozen and the combinatorics is set free, so a geometry becomes a gluing pattern of simplices whose shapes are fixed once and for all, and summing over geometries becomes counting gluings.

In both cases the geometry is carried by the gluing and not by any ambient space. Each simplex is flat inside, so curvature can only sit where simplices meet, and it is measured by how far the angles around such a meeting place fall short of a full turn. Figure ?? shows the three cases for triangles glued around a vertex.

Because the combinatorics is now free, the number of simplices differs from history to history, and within a single history the number of simplices in a slice changes from one slice to the next.

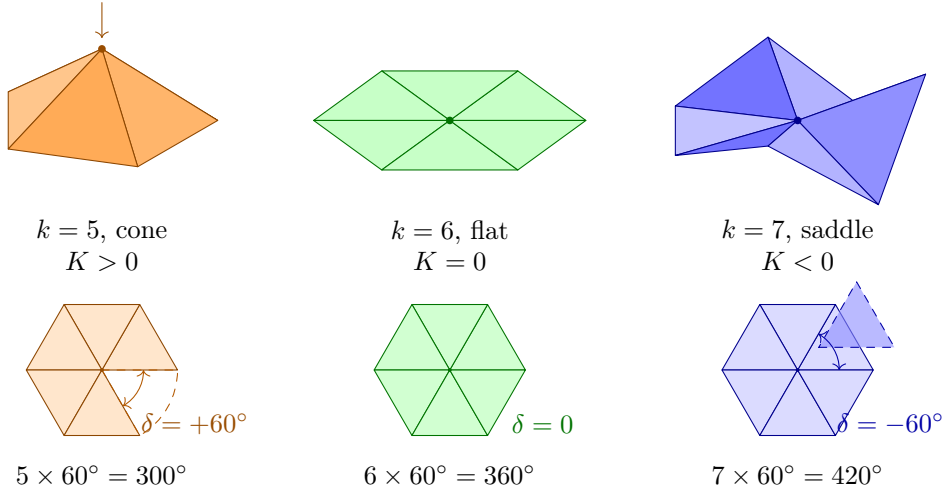


Figure 1: Curvature in Regge calculus lives at the vertices, not on the interiors of the simplices. Gluing k flat unit equilateral triangles around a single point makes the angles there sum to $k \cdot 60^\circ$, and the deficit angle $\delta = 2\pi - \sum_i \theta_i$ measures the failure to close up in the plane. Top row shows the resulting surface in $\mathbb{R}^3$, bottom row shows the same triangles unfolded flat, where δ is directly visible. For $k = 5$ the angles fall short, the gap closes only by lifting the vertex out of the plane, and the cone carries positive curvature concentrated at a point. For $k = 6$ the angles close exactly and the neighbourhood is isometric to a patch of the Euclidean plane. For $k = 7$ the seventh triangle (dashed) overlaps the first, the excess forces the surface to buckle into a saddle, and the curvature is negative. Every triangle is flat throughout, so all the geometry is carried by the combinatorics of how they are glued.

New geometric measure appears from nowhere. Classical general relativity is no better off on this point, since it has no conservation law for spatial volume either, and in a closed universe the Hamiltonian constraint balances the total at zero [4, 9] however much volume appears. The complaint is therefore not that volume grows, but that the growth answers to nothing the theory conserves. A cosmological constant term only assigns a cost to four-volume within the ensemble, and where the total is held fixed instead, as in simulations and in unimodular gravity [15, 11], the constraint is imposed by hand rather than descending from a symmetry.

This paper takes the discomfort seriously and asks what a theory looks like if one demands, from the start, that

1. Nothing is ever destroyed. Evolution proceeds only by forking and completion moves, never by deletion, so that topological invariants are conserved by construction.
2. Time is not a dimension of the fundamental structure but an emergent bookkeeping of an irreversible forking-filling process.
3. Entropy is attached to frame-invariant causal data, not to spatial slices, so that the theory never needs to say what all of space looks like at a single moment.
4. The rule that chooses which move happens next is derived from an inference principle rather than postulated.

In our framework the universe begins as a single vertex, which we call the big bang, and grows by two kinds of local moves:

1. A **sprout** attaches a fresh directed edge ending at a new vertex, adding new space.

2. A **fill** completes an unfilled inner horn, attaching a composite edge together with the higher simplex that witnesses the composition.

Two kinds of direction appear here. Every edge of the simplicial set carries an orientation, and that orientation is what decides which horns count as inner and what composition means. Separately, every move is an event, and the events are ordered by which of them created the simplices the others apply to. We call the first the **orientation** and the second the **causal order**. No move ever deletes anything, so growth always runs forward in the causal order, while orientation is a property of the finished simplices and can be reversed on paper.

At the big bang two edges sprout from the single vertex with opposite orientation, and this is what produces the two branches. The mirror branch is not an independent run. By definition it is the edge-reversal of the branch, carrying one mirrored event for every event, which is what makes the pair a mirror image of itself. The mirroring is causally inert, since no move in one branch ever applies to a simplex created in the other, so the causal past of any event lies wholly inside its own branch. Each branch grows forward in its own causal order while pointing the opposite way in orientation, so neither branch runs backward in time in any absolute sense. Each is backward only as seen from the other, and there is no global causal order in which either of them is reversed. This is a combinatorial counterpart of CPT-symmetric cosmologies, where a universe and an antiuniverse emerge together from a single point [6, 5].

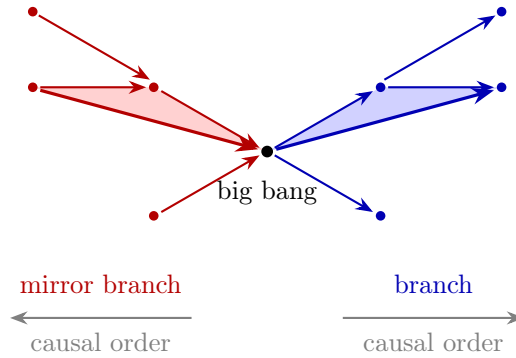


Figure 2: The total object. A branch on the right and its mirror image on the left, glued at the single vertex where both were born. Corresponding edges carry opposite orientation, so the pair is isomorphic to its own edge-reversal. The shaded region on each side is a filled inner horn, drawn with its composite edge in bold, and the fill is mirrored along with everything else. The grey arrows are the causal order, which runs away from the shared vertex on both sides, so each branch is backward only as seen from the other, and never in the order in which its own simplices were built.

At any moment the state of the universe is a finite simplicial set with directed edges, together with the history of how it was built. The limit of the filling process in one branch is a quasicategory, in which every composition has been formed while the orientations remain. The classical, zero-information limit is the nerve of an ordinary category, the simplicial set whose n -simplices are the chains of n composable morphisms [14, 12], and in it every filler is unique. The physical content of a completed universe is precisely how far it is from being a nerve.

The pair of branches is meant to carry no information at all. Two contractible halves glued at a single vertex are again contractible, so every homotopy invariant of the pair is that of a point, and the pair is isomorphic to its own edge-reversal, so it carries no arrow either. Whatever is physical therefore sits inside a single branch, in the orientations and in the history of how its simplices were built.

2 Preliminaries and related work

3 The framework

3.1 The object

At any moment, the universe is a finite simplicial set. Its edges are directed by the simplicial structure itself, since the face maps assign every edge a source vertex and a target vertex. It begins as a single vertex Δ^0 and it grows from there.

The object is pre-geometric. It is not spacetime, and it is not a representation of a spacetime lying elsewhere. It carries no metric, no dimension, and no manifold structure. Everything geometric has to be recovered from it. Statements about space are made in the framework's own combinatorial terms.

3.1.1 Growing the Simplicial Set

Definition 3.1 (Move). The simplicial set grows by two local, non-deleting **moves**, which are pushouts along the following two monomorphisms of finite simplicial sets, where $\hookrightarrow$ denotes an inclusion of one simplicial set into another:

1. **Sprout:** the coface $d^1 : \Delta^0 \hookrightarrow \Delta^1$, including Δ^0 as the source vertex. A sprout can happen at any vertex v of the current state. Performing it attaches a fresh directed edge from v to a new vertex.
2. **Fill:** $\Lambda_k^n \hookrightarrow \Delta^n$ for $0 < k < n$. A fill can happen at any unfilled inner horn. Performing it attaches the missing face and interior. Each horn may be filled at most once.

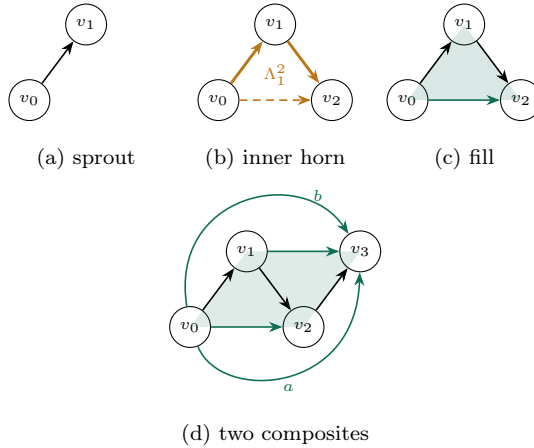


Figure 3: The two moves and what each of them adds. In (a) a vertex sprouts a fresh directed edge to a new vertex. After a second sprout, (b) shows the composable pair $v_0 \rightarrow v_1 \rightarrow v_2$ whose composite is absent, an unfilled inner horn Λ_1^2 (amber, missing edge dashed). In (c) the fill supplies the composite $v_0 \rightarrow v_2$ (green) together with the 2-simplex witnessing it (shaded). Panel (d) shows the state after a third sprout and both first-tier fills. Two distinct horns now call for a composite from v_0 to v_3 , one built from $v_0 \rightarrow v_1$ with $v_1 \rightarrow v_3$ and one from $v_0 \rightarrow v_2$ with $v_2 \rightarrow v_3$. Each fill mints its own composite, so a and b are parallel edges rather than a single shared one. Only once both are present does the inner horn Λ_1^3 exist, since its three faces already carry all six edges of Δ^3 . Filling it therefore adds a 2-simplex and a 3-simplex but no edge. Only inner horns are ever filled, so edge direction is never inverted and the limit object is a quasicategory.

The initial object is a single vertex Δ^0 . At any moment the current state is a finite simplicial set, and the next move is chosen from the set of available moves, which is the disjoint union of all sprout sites and all unfilled inner horns.

Every vertex is created by exactly one sprout, which supplies its first incoming edge. Every other incoming edge is a composite supplied by a fill. So sprout edges and composite edges never mix. A sprout creates a fresh vertex and attaches a directed edge to it. The composite joins two vertices that already exist.

Sprouting is a Whitehead elementary expansion and inner-horn filling is an anodyne extension. Both operations preserve homotopy equivalences, so the homotopy type of the simplicial set is conserved by construction. In our case, the homotopy type remains that of a point.

Which moves add an edge is a combinatorial fact about horns.

Lemma 3.2 (Which moves add edges). *A sprout adds exactly one edge, a fill in dimension two adds exactly one edge, and a fill in dimension three or above adds no edge.*

Proof. A sprout attaches Δ^1 along its source vertex, so it adds the one edge of Δ^1 . An inner horn Λ_k^n contains every face of Δ^n except the interior and the k -th $(n-1)$ -face. For $n=2$ the missing face is itself an edge, the composite, and the fill adds exactly that edge. For $n \geq 3$, an edge $\{i, j\}$ of Δ^n lies in the face opposite vertex l for every $l \notin \{i, j\}$, and at least two such l exist because Δ^n has at least four vertices. The horn misses at most one face, so every edge of the attached Δ^n already lies in a face the horn contains, and the fill adds no edge. $\square$

Fills in dimension three and above add only simplices that say existing paths agree. So every chain is built out of edges, and nothing above dimension one can be part of a chain. Whatever geometric and metric structure this model has is carried by the vertices, the edges, and the causal order between events. Treat the edges as a multigraph. If two fills each produce a composite, keep both as separate edges. The number of parallel edges is a record of how many fills happened there. The order says which events can affect which. The edge counts say how much happened. Both are needed, and neither replaces the other.

Definition 3.3 (Rule). The **rule** of an event is a specification that contains three pieces of information:

1. **Find**: Which spots in the current simplicial set qualify for the move to happen.
2. **Match**: Which spot the move is applied to.
3. **Apply**: What new simplices the move puts in there.

Formally, the rule is an inclusion $F \hookrightarrow A$ of finite simplicial sets, from the **find pattern** F to the **apply pattern** A . The match is an injective map $m_e : F \hookrightarrow X_{<e}$ from the find pattern into the state of the events strictly preceding e (Definition 3.11), and apply is the pushout of the rule along the match. Injectivity forbids folded spots, where two simplices of the pattern land on the same simplex of the state. The event adds the simplices of A not in the image of F . The find and apply patterns of an event e are written F_e and A_e .

Three kinds of rules exist:

- **The big bang.** Find: nothing needs to be found. Match: there is no spot to choose. Apply: the initial vertex Δ^0 is created out of nothing ($\emptyset \hookrightarrow \Delta^0$).
- **A sprout.** Find: every vertex qualifies. Match: the chosen vertex v , a map $\Delta^0 \rightarrow X_{<e}$. Apply: attach Δ^1 by its source vertex (the coface $d^1 : \Delta^0 \hookrightarrow \Delta^1$), adding an edge from v to a fresh vertex.

- **A fill.** Find: every unfilled inner horn Λ_k^n with $0 < k < n$ qualifies. Match: the chosen horn, an embedded copy of Λ_k^n in $X_{<e}$. Apply: attach Δ^n along the horn ($\Lambda_k^n \hookrightarrow \Delta^n$), adding the missing face and the interior.

Remark 3.4 (The at-most-once convention). The last clause of Definition 3.1 is automatic under the fill rule as stated, since the find pattern offers only unfilled inner horns, and a horn, once filled, is never offered again. The clause is kept because it records a choice. The alternative convention lets a filled horn be filled again, so that one horn can supply several witnesses to the same composite. That variant changes what the events record, and with it any entropy built from the event record. It is not studied here.

3.2 History, runs, and the causal order

Growth happens one move at a time. The universe at any moment is a finite stage of an ongoing history, and each event after the big bang is a move applied at a particular spot. Since no move deletes anything, an event can only apply to simplices that already exist. Event b therefore depends on event a when the spot at which b is applied contains, among its maximal simplices, one created by a , where a simplex of the spot is maximal when it is not a face of another simplex of the spot. For a sprout the spot is a single vertex, which is maximal. For a fill the maximal simplices of the horn Λ_k^n are its $(n-1)$ -faces, so a dimension-two fill depends on the two events that created the edges of its horn and not directly on the events that created its vertices. The vertex creators still lie in the causal past by transitivity, since a simplex is only ever created after its vertices already exist. The causal order of an event e is the set of events that e depends on directly or via other events, together with e itself, with the order among them. A reference frame is a linear extension of this order [7].

Definition 3.5 (History). A **history** $\mathcal{H}$ is a partially ordered set of events with a least element $\perp$, the big bang, in which every other event e is a sprout or a fill applied at a specified spot of the simplicial set built by the events strictly preceding e , subject to three conditions:

1. Every event has only finitely many events preceding it.
2. The big bang creates the initial vertex Δ^0 .
3. The order is generated by creation-before-use.

A history maintains a record of every event and all of its data, namely its rule. Different histories can result in the same simplicial set. Nothing in the framework as stated selects one history over another, and we do not treat the collection of all histories as a single mathematical object here. What selects the realized history is left open. Posing that selection as a well-formed inference problem is one of the aims of this work.

Definition 3.6 (Causal Order). A **causal order** $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ of a history $\mathcal{H}$ for an event e is the set of events preceding and including e , partially ordered by $\prec$, where $a \prec b$ if and only if b depends on a directly or via other events. Direct dependency is witnessed only by the maximal simplices of the spot, as fixed above, so a fill depends directly on the creators of the faces of its horn and only indirectly on the creators of its vertices. The **causal past** of e is the causal order with e itself removed.

Lemma 3.7 (Acyclicity of the one-skeleton). *The directed multigraph formed by the vertices and edges of the state at any event has no directed cycles.*

Proof. Every vertex is created by a unique event, the big bang for the initial vertex and a sprout for every other vertex, and we write $\varepsilon(v)$ for the event that created v . We show by induction

along the causal order that every edge $u \rightarrow w$ of every state satisfies $\varepsilon(u) \prec \varepsilon(w)$. A sprout edge leaves a vertex that already exists and ends at the fresh vertex the sprout creates, so the creator of its source lies strictly below the sprout, which is the creator of its target. A composite edge $u \rightarrow w$ supplied by a fill comes with the two edges $u \rightarrow v$ and $v \rightarrow w$ of its horn, which were created earlier, and induction gives $\varepsilon(u) \prec \varepsilon(v) \prec \varepsilon(w)$. By Lemma 3.2 these are all the edges there are. A directed cycle $v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_0$ would now force $\varepsilon(v_0) \prec \varepsilon(v_0)$, which a strict partial order forbids. $\square$

Definition 3.8 (Run). The **run** at event e , written $\mathcal{R}(e)$, is the subposet of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ consisting of the big bang and the events that add an edge, with the induced order. By Lemma 3.2 these are exactly the sprouts and the dimension-two fills.

The run is closed downward. A sprout depends only on the event that created its vertex, and vertices are created only by sprouts. A dimension-two fill depends only on the events that created the two edges of its horn, and edges are created only by sprouts and dimension-two fills. So no edge-adding event ever depends on a higher fill, and the run at e is a downset [7] of the history. A general downset of the history is not a run: downsets may contain higher fills, while the run keeps only the edge-adding events. The induced order is automatically a partial order, since a partial order restricted to a subset is still a partial order. Moreover, the run at e is a history in its own right, since it satisfies the three conditions:

1. Every event of the run has only finitely many events below it in the history, hence also in the run.
2. The big bang is included by definition and stays the least event.
3. The induced order on the run is generated by creation-before-use. A sprout depends only on the event that created its vertex, which is a sprout, and a dimension-two fill depends only on the events that created the two edges of its horn, which are sprouts or dimension-two fills. So every direct dependency of an edge-adding event is again an edge-adding event, and dependency chains between run events never leave the run.

At any moment, the run is the partially ordered record of exactly those events that change the one-skeleton. It determines the one-skeleton of the state as a multigraph, with parallel composites kept distinct, and it forgets the higher fills entirely. Passing from a history to its run forgets the higher fills, and passing from a run to its one-skeleton forgets the order. Different histories can produce the same run, since higher fills leave no trace in it. Different runs can produce the same one-skeleton, since the multigraph forgets the order. Everything metric in this framework will be a function of the run, while the entropy of an event is a function of its causal order, which remembers the higher fills that the run forgets.

Definition 3.9 (Entropy). For a finite poset P , write $\mathcal{L}(P)$ for its set of linear extensions. The **entropy** of an event e is

$$S(e) = \log |\mathcal{L}(\mathcal{C}_{\prec_{\mathcal{H}}}(e))|,$$

the logarithm of the number of linear extensions of the causal order. The event e is the unique maximum of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$, so the causal past of e has the same number of linear extensions, and the entropy measures how much of the past could have happened in a different order. The logarithm is natural.

Example 3.10. Start from the single vertex v_0 , created by the big bang. The history grows by the following events.

1. s_1 : sprout an edge $v_0 \rightarrow v_1$.

2. s_2 : sprout an edge $v_1 \rightarrow v_2$.
3. s_3 : sprout a second edge $v_0 \rightarrow v_3$ from the starting vertex.
4. f : fill the horn $v_0 \rightarrow v_1 \rightarrow v_2$, supplying the composite $v_0 \rightarrow v_2$ and its witnessing 2-simplex.
5. s_4 : sprout an edge $v_2 \rightarrow v_4$.
6. g : fill the horn $v_0 \rightarrow v_2 \rightarrow v_4$, whose first edge is the composite from f , supplying the composite $v_0 \rightarrow v_4$.
7. f' : fill the horn $v_1 \rightarrow v_2 \rightarrow v_4$, supplying the composite $v_1 \rightarrow v_4$. Now all six edges among v_0, v_1, v_2, v_4 exist, and three of the four triangular faces carry 2-simplices.
8. h : fill the resulting inner horn Λ_2^3 , supplying the missing face (v_0, v_1, v_4) and the 3-simplex on all four vertices, and adding no edge.

The causal order is generated by $s_1 \prec s_2 \prec f \prec g \prec h$ and $s_2 \prec s_4 \prec g, s_4 \prec f' \prec h$, while s_3 shares only the initial vertex with the others and stays incomparable to all of them. The causal order of g is $\mathcal{C}_{\prec_{\mathcal{H}}}(g) = \{s_1, s_2, f, s_4, g\}$, in which f and s_4 are incomparable, so $|\mathcal{L}(\mathcal{C}_{\prec_{\mathcal{H}}}(g))| = 2$ and $S(g) = \log 2$. We suppress the big bang $\perp$ from the notation for causal orders, since it sits below every event.

The three objects of this section differ at h . The causal order is $\mathcal{C}_{\prec_{\mathcal{H}}}(h) = \{s_1, s_2, s_4, f, g, f', h\}$, with 5 linear extensions, so $S(h) = \log 5$. The run at h is the same poset with h removed, since h adds no edge and leaves no trace in the edge record. The state at h is the simplicial set on v_0, v_1, v_2, v_4 with all six edges, all four 2-simplices, and the 3-simplex, carrying no event labels, and it does not contain v_3 , since s_3 is not in the causal order of h .

Definition 3.11 (State). The **state** at e , written X_e , is the colimit of the diagram of rule applications in $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$. The diagram is indexed by the downsets of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$, sending each downset D to the simplicial set X_D built by the events of D and each inclusion of downsets to the induced inclusion of states. We write $X_{<e}$ for $X_{\mathcal{C}_{\prec_{\mathcal{H}}}(e) \setminus \{e\}}$, the state of the causal past, on which the event e acts.

The colimit is used here so that ordering of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ has no consequence on the resulting state. Whether the diagram is well-defined, and whether it agrees with what any sequential build produces, is the content of Proposition 3.13.

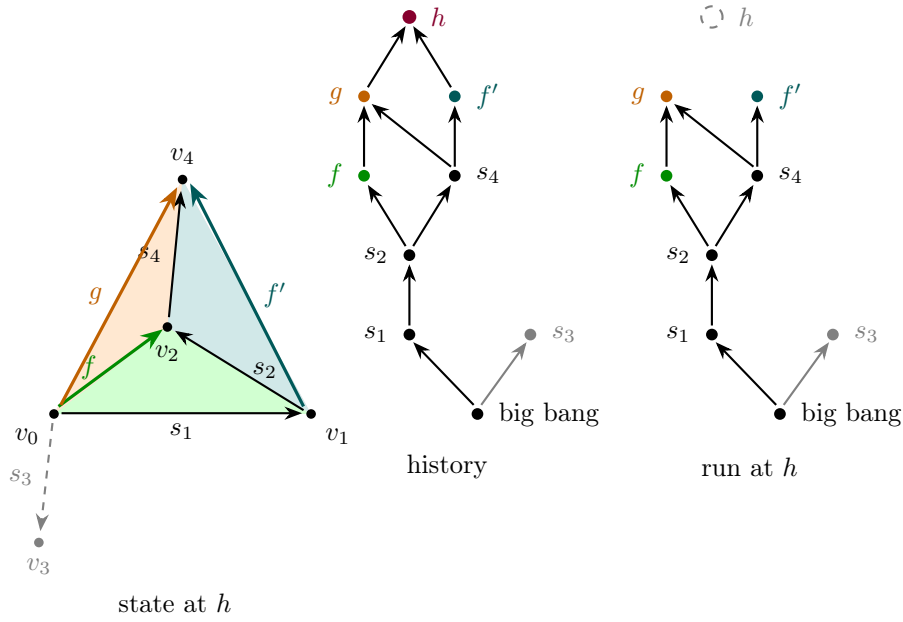


Figure 4: Example 3.10. Left: the state at h , the simplicial set itself, with sprout edges plain and the three composites colored by the fill that supplied them. The face (v_0, v_1, v_4) (purple) and the solid tetrahedron come from h . Middle: the history, with h (purple) a dimension-three fill sitting above the fills it depends on. Right: the run at h , the same order with h forgotten, since h adds no edge. The history remembers every event, the run drops the higher fill, and the state drops the order and the labels.

Lemma 3.12 (Exchange of order). *Let X be a simplicial set and let a and b be two events whose matches both factor through X . Then a and b can be applied in either order, and the two resulting simplicial sets are canonically isomorphic.*

Proof. Pushouts of monomorphisms in $\mathbf{sSet}$ are monomorphisms, so the attachment map $X \rightarrow X \cup_{F_a} A_a$ is a monomorphism, and since no simplices are deleted or re-glued, the match of b persists as a map into $X \cup_{F_a} A_a$. The pushout property used here holds because pushouts in $\mathbf{sSet}$ are computed degree by degree, and in the category of sets the pushout of an injection along any map is an injection, which is the adhesiveness property of presheaf categories [10]. Symmetrically, a can be applied after b :

$$\begin{array}{ccc}
 & X \cup_{F_a} A_a & \\
 a \nearrow & & \searrow b \\
 X & & X_{ab} \\
 b \searrow & & \nearrow a \\
 & X \cup_{F_b} A_b &
 \end{array}$$

The two routes compute the colimit of the same diagram, formed by X , the two matches, and the two rules, so the resulting states X_{ab} are canonically isomorphic by the pasting law for pushouts. This is the local Church–Rosser property for non-deleting double-pushout rewriting [10]. $\square$

Proposition 3.13 (Well-definedness of the state). *The causal order on the events of a history is a partial order, and for every event e the colimit in Definition 3.11 exists and agrees with the state produced by any sequential build along a linear extension of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$.*

Proof. There are three claims to prove:

1. The causal order $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ of an event e is a partial order.
2. The states X_D exist for all finite downsets D of $\mathcal{H}$.
3. The state at e agrees with every sequential build of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$.

We prove them in that order.

Claim 1: the causal order is a partial order. This claim is a consistency check on Definition 3.5, not a theorem about an unordered growth process. A history presents its events as a partially ordered set from the outset, and the third condition asks this order to be generated by creation-before-use. What needs checking is that the condition can be met, meaning that the dependency relation generates a strict partial order. Recall that b depends on a when a maximal simplex of the spot of b is created by a . The spot of b lies in the simplicial set built by the events strictly below b , so every simplex of the spot is created by an event strictly below b , and dependency never points upward. The transitive closure of the dependency relation is therefore contained in the strict order of the history, and a cycle of dependencies would be a cycle in that strict order, which has none. So the dependency relation generates a strict partial order on $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$, as the third condition requires. A stronger formulation is possible, in which events are given as a bare sequence of move applications with no order posited in advance, and dependency is shown acyclic by induction on position in the sequence. We do not develop that presentation here.

Claim 2: the state of a finite downset exists. For a finite downset $D \subseteq \mathcal{H}$ we construct the simplicial set X_D by induction on $|D|$, the number of events in D . The induction is an algorithm that removes one event per step.

1. *Base case.* If $D = \emptyset$, set $X_{\emptyset} = \emptyset$, the empty simplicial set.
2. *Inductive step.* Suppose $X_{D'}$ has been constructed for every downset D' with $|D'| < |D|$. Choose a maximal element b of D and set $D' = D \setminus \{b\}$, which is again a downset and has one element fewer. Let $F_b \hookrightarrow A_b$ be the rule of b and $m_b : F_b \rightarrow X_{\prec_b}$ its match, as in Definition 3.3. Every simplex in the image of m_b is created by an event strictly below b , and every such event lies in D' , so the image sits inside $X_{D'}$ with its face relations unchanged, since no move deletes or re-glues anything. We therefore also write m_b for the map $F_b \rightarrow X_{D'}$. Now define X_D by the pushout square

$$\begin{array}{ccc} F_b & \hookrightarrow & A_b \\ m_b \downarrow & & \downarrow \\ X_{D'} & \longrightarrow & X_D \end{array}$$

which attaches the new simplices of A_b to $X_{D'}$ along the spot m_b .

3. *Termination.* Each step removes one event, so the recursion reaches $\emptyset$ after exactly $|D|$ steps. Since D is finite, this always stops.
4. *The choice of maximal element does not matter.* If $b_1 \neq b_2$ are both maximal in D , they are incomparable, and $D'' = D \setminus \{b_1, b_2\}$ is again a downset, and it contains every event strictly below b_1 or b_2 . Both matches therefore factor through $X_{D''}$. By the induction hypothesis, $X_{D \setminus \{b_1\}}$ can be computed by attaching b_2 to $X_{D''}$, and $X_{D \setminus \{b_2\}}$ can be computed by attaching b_1 to $X_{D''}$. Lemma 3.12 identifies the two routes to X_D , and induction on $|D|$ then shows that X_D is independent of all such choices.

Applying this to $D = \mathcal{C}_{\prec_{\mathcal{H}}}(e)$ defines the state $X_e := X_{\mathcal{C}_{\prec_{\mathcal{H}}}(e)}$. The causal order $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ is finite by the definition of a history, so the algorithm applies.

Claim 3: every sequential build agrees with the colimit. Fix a linear extension of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ and build the state event by event along it.

1. Any two linear extensions of a finite poset are connected by a sequence of adjacent transpositions of incomparable elements. If a and b are adjacent in the extension and incomparable in the causal order, then every event strictly below b precedes a in the extension, since it cannot lie between two adjacent events and cannot be a itself. Both matches then factor through the state built before a , and Lemma 3.12 shows that swapping the two events does not change the result. So all sequential builds of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$ agree.

2. The chain of downsets

$$\emptyset \subsetneq \{e_1\} \subsetneq \{e_1, e_2\} \subsetneq \cdots \subsetneq \mathcal{C}_{\prec_{\mathcal{H}}}(e)$$

built along a linear extension is final in the poset of downsets of $\mathcal{C}_{\prec_{\mathcal{H}}}(e)$, meaning that every downset is contained in some member of the chain, and the members are nested. Hence the colimit over the downset diagram equals the colimit along the chain.

3. The colimit along the chain is the iterated pushout, which is exactly what the sequential build computes.

Hence every sequential build produces the state X_e of Definition 3.11. □

4 Results

5 Discussion

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